

EXPERIMENT No. - 42083/
Date: 02.12.6**TO DETECT THE HETERO ELEMENT OR FOREIGN ELEMENT PRESENT IN THE GIVEN ORGANIC COMPOUND. (Sample E₁)****APPARATUS REQUIRED**

- | | |
|-----------------|---------------------|
| a) Test tube | b) fusion tube |
| c) funnel | d) Test tube holder |
| e) filter paper | f) Mortar & Pestle |
| g) Burner | h) Glass rod |
| i) | j) |

CHEMICAL REQUIRED

- | | |
|---|--|
| a) A piece of Sodium metal | b) Given organic compound |
| c) NaOH | d) FeSO ₄ |
| e) FeCl ₃ | f) HCl |
| g) Na ₂ [Fe(CN) ₅ NO] | h) (CH ₃ COO) ₂ Pb |
| i) CH ₃ COOH | j) AgNO ₃ |

THEORY

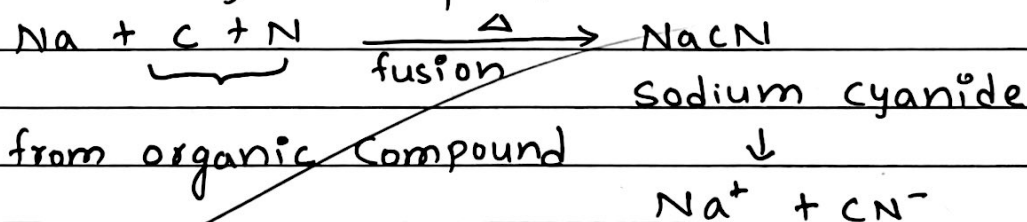
Hydrocarbon & their derivatives, in which covalently bonded carbon is the essential element are called organic compounds. For eg: ethanol (C₂H₅OH), aniline (C₆H₅NH₂), Urea (NH₂CONH₂), ethanoic acid (CH₃COOH) etc. All organic compound contain carbon. They are no organic compound without carbon. For e.g: CCl₄, C₂Cl₆ etc. Hence parent organic compounds are hydrocarbons. Some organic compounds contain other element like oxygen, nitrogen, sulphur, halogen, phosphorous & some metals.

Hence, elements other than carbon & hydrogen present in organic compounds are called Hetero-element or Foreign element.

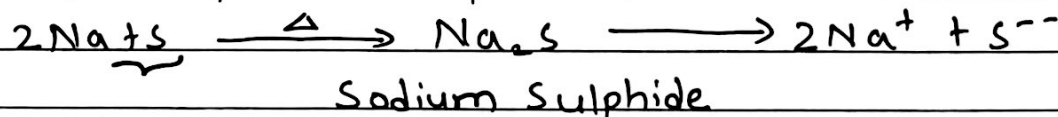
Organic compounds are covalent in nature. Foreign element present in organic compound are also bonded covalently. Therefore, the laboratory reagent do not directly react with these compounds. Hence, first these element are converted from covalent to ionic compounds for this purpose sodium extract should be prepared.

The sodium fusion test for foreign elements was performed by french J.L. Lassaigne in 1843 and is known as lassaigne test or sodium metal following reactions take place

1) If nitrogen is present

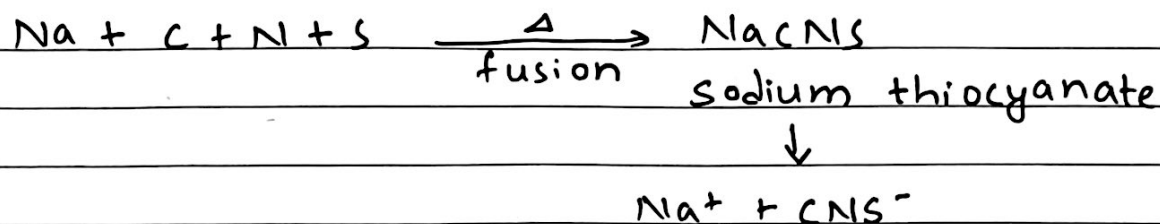


2) If sulphur is present

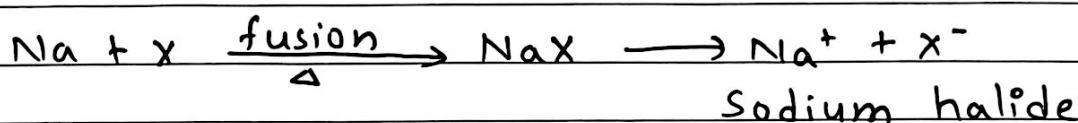


from organic compound

3) If both nitrogen & sulphur are present together.



4) If halogen is present



PROCEDURE

A freshly cut sodium metal above the size of grain rice was taken. The piece of sodium metal was dried by pressing between the fold of filter paper and was held with the help of tongs & was heated gently till the sodium metal fuse & form shining globule. The fusion tube was taken out of the flame and was cooled then small amount of organic compound was added. Fusion tube was again heated first gently & then ~~sin~~ strongly until the entire nearer end of fusion tube become red hot. The red hot tube was plugged in 10ml distilled water taken in a porcelain basin or mortar with the help of pestle. The content was boiled for few minutes & was filtered. The filtrate was called sodium extract & this solution is used to perform following tests.

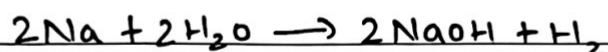
OBSERVATION TABLE

S.N.	Experiments	Observations	Inferences
1.	Detection of nitrogen 1 ml of sodium extract was taken in a clean test tube, few drops of NaOH was added & then 1 ml of freshly prepared ferrous sulphate soln was added. The mixture was boiled with shaking then cooled it. 1 to 2 drops of FeCl_3 soln was added & acidified with Conc. HCl.	1.5 ml of prussian blue coloration was appeared.	presence of nitrogen
2.	Detection of sulphur a) Sodium nitroprusside test 1 ml of sodium extract was taken in a clean test tube, few drops of sodium nitroprusside solution was added.	No violet coloration was appeared.	Absence of Sulphur

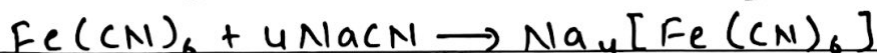
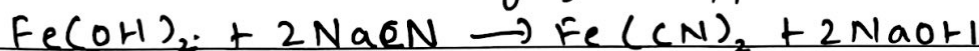
	b) Lead acetate test 1 ml of sodium extract was taken in test tube. It was acidified with few drops of lead acetate was added.	No black ppt was appeared.	Absence of sulphur
3.	Detection of both nitrogen and sulphur together 1 ml of sodium extract was taken in a test tube & few drops of FeCl_3 were added.	No blood red colouration was appeared.	Absence of both nitrogen & sulphur together.
4.	Detection of halogen 1 ml of sodium extract was taken in a test tube & few drops of dil. HNO_3 was added. The content was ad boiled, cooled & few drops of AgNO_3 were added.	No ppt was appeared.	Absence of halogen

TEST REACTIONS

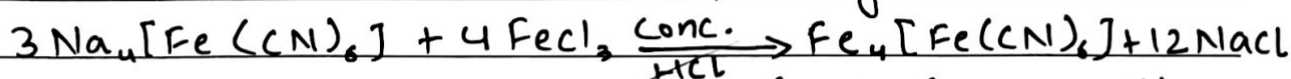
• Detection of Nitrogen



dirty green ppt.



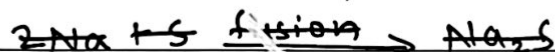
Sodium ferrocyanide



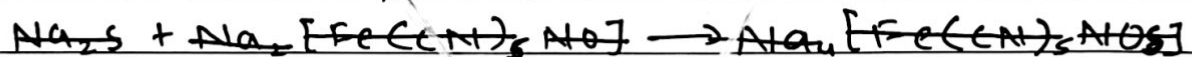
ferric ferrocyanide

(Prussian blue or green color)

• Detection of sulphur



i) Sodium nitroprusside test

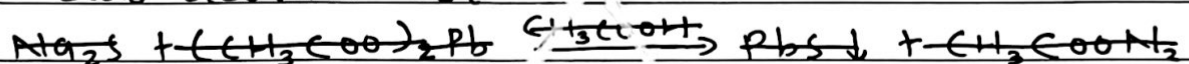


Sodium nitroprusside

Sodium sulphonitro

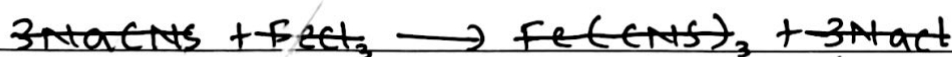
prusside (violet color)

ii) Lead acetate test



(black ppt)

• Detection of nitrogen & sulphur together:



RESULT

In this way the foreign element present in given organic compound was detected to be nitrogen.

CONCLUSION

Hence, from above experiment it can be concluded that the foreign element present in the organic compound can be detected by using Lassaigne's test.

PRECAUTIONS

- 1) Personal protective equipments should be worn.
- 2) Sodium extract should be prepared in the distilled water.
- 3) Apparatus should be handled with care.

VIVA-VOCE

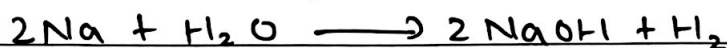
1. What is meant by foreign element?
2. Why is sodium fused with organic compound?
3. Is potassium be used in place of sodium?
4. Why is sodium extract alkaline in nature?
5. Why is it essential to boil sodium extract with nitric acid before halogen test?

1) Element other than carbon & hydrogen present in organic compound are called foreign element.

2) The foreign element in organic compound are covalently bonded to carbon atom and reaction are slow. So, they are fused with sodium to convert covalent bond to ionic bond.

3) No, potassium can't be used because it forms poisonous KCN with organic compound & potassium is highly reactive than sodium & reaction is violent.

4) If sodium isn't completely ~~removed~~ reacted with organic compound during fusion, unreacted sodium hydroxide is formed with water.



5) Sodium extract should be boiled with nitric acid to remove pseudohalogen (CN^-), (S^{--}) which may interfere the halogen test.

[Signature]
03/23